

EXPERIMENT 3 — NESTED CROSS-VALIDATION RESULTS

To address the reviewers’ requests for a more rigorous statistical evaluation, we report below the repeated stratified subject-level evaluation underlying our nested cross-validation protocol (5 outer folds $\times$ 5 repeats = 25 outer train/test partitions; an inner loop, nested strictly within each outer training partition, was used for hyperparameter search only) for both the HIS-RKHS (MMD²) and PPK kernel families, each combined with KNN and SVM classifiers, across the three HMM topologies evaluated (N3G3, N4G4, N5G5 = number of hidden states $\times$ number of GMM components per state). Hyperparameters were selected exclusively from the training partition of each outer fold, preventing information leakage from the held-out test fold into model selection. For every configuration we report bootstrap 95% confidence intervals CI for accuracy, balanced accuracy, sensitivity, specificity, AUC-ROC, and the Matthews correlation coefficient (MCC), the pooled confusion matrix over the 25 outer folds, and a label-permutation significance test ($n = 1000$ permutations).

Dataset N3G3

(HIS-RKHS)

KNN

Table 1: Repeated stratified subject-level evaluation (5 folds $\times$ 5 repeats = 25 train/test partitions), bootstrap 95% CI ($n = 2000$) — (HIS-RKHS) – KNN, dataset N3G3.

Metric	Mean	CI Lower	CI Upper
Accuracy	0.6642	0.6304	0.6996
Balanced Accuracy	0.6633	0.6296	0.6984
Sensitivity	0.7067	0.6485	0.7610
Specificity	0.6200	0.5633	0.6767
AUC-ROC	0.7314	0.6863	0.7737
MCC	0.3388	0.2684	0.4108

Table 2: Pooled confusion matrix over the 25 outer test folds and label-permutation test ($n = 1000$ permutations) — (HIS-RKHS) – KNN, dataset N3G3. Permutation p -value: <0.001 .

	Pred. Control	Pred. ADHD
True Control	186	114
True ADHD	89	216

SVM

Table 3: Repeated stratified subject-level evaluation (5 folds $\times$ 5 repeats = 25 train/test partitions), bootstrap 95% CI ($n = 2000$) — (HIS-RKHS) – SVM, dataset N3G3.

Metric	Mean	CI Lower	CI Upper
Accuracy	0.7353	0.6977	0.7699
Balanced Accuracy	0.7350	0.6978	0.7696
Sensitivity	0.7733	0.7282	0.8203
Specificity	0.6967	0.6233	0.7667
AUC-ROC	0.7960	0.7640	0.8279
MCC	0.4828	0.4063	0.5518

Table 4: Pooled confusion matrix over the 25 outer test folds and label-permutation test ($n = 1000$ permutations) — (HIS-RKHS) – SVM, dataset N3G3. Permutation p -value: <0.001 .

	Pred. Control	Pred. ADHD
True Control	209	91
True ADHD	69	236

Probability Product Kernel (PPK)

PPK-KNN

Table 5: Repeated stratified subject-level evaluation (5 folds $\times$ 5 repeats = 25 train/test partitions), bootstrap 95% CI ($n = 2000$) — Probability Product Kernel (PPK) – PPK-KNN, dataset N3G3.

Metric	Mean	CI Lower	CI Upper
Accuracy	0.7235	0.6821	0.7640
Balanced Accuracy	0.7229	0.6819	0.7635
Sensitivity	0.7992	0.7556	0.8410
Specificity	0.6467	0.5900	0.7033
AUC-ROC	0.7649	0.7306	0.7992
MCC	0.4555	0.3722	0.5368

Table 6: Pooled confusion matrix over the 25 outer test folds and label-permutation test ($n = 1000$ permutations) — Probability Product Kernel (PPK) – PPK-KNN, dataset N3G3. Permutation p -value: <0.001 .

	Pred. Control	Pred. ADHD
True Control	194	106
True ADHD	61	244

PPK-SVM

Table 7: Repeated stratified subject-level evaluation (5 folds $\times$ 5 repeats = 25 train/test partitions), bootstrap 95% CI ($n = 2000$) — Probability Product Kernel (PPK) – PPK-SVM, dataset N3G3.

Metric	Mean	CI Lower	CI Upper
Accuracy	0.7061	0.6671	0.7467
Balanced Accuracy	0.7055	0.6664	0.7456
Sensitivity	0.7677	0.7126	0.8241
Specificity	0.6433	0.5900	0.7000
AUC-ROC	0.7624	0.7012	0.8111
MCC	0.4245	0.3446	0.5080

Table 8: Pooled confusion matrix over the 25 outer test folds and label-permutation test ($n = 1000$ permutations) — Probability Product Kernel (PPK) – PPK-SVM, dataset N3G3. Permutation p -value: <0.001 .

	Pred. Control	Pred. ADHD
True Control	193	107
True ADHD	71	234

Dataset N4G4

(HIS-RKHS)

KNN

Table 9: Repeated stratified subject-level evaluation (5 folds $\times$ 5 repeats = 25 train/test partitions), bootstrap 95% CI ($n = 2000$) — (HIS-RKHS) – KNN, dataset N4G4.

Metric	Mean	CI Lower	CI Upper
Accuracy	0.6855	0.6436	0.7248
Balanced Accuracy	0.6846	0.6429	0.7236
Sensitivity	0.7526	0.6943	0.8095
Specificity	0.6167	0.5467	0.6833
AUC-ROC	0.7642	0.7218	0.8031
MCC	0.3863	0.3017	0.4668

Table 10: Pooled confusion matrix over the 25 outer test folds and label-permutation test ($n = 1000$ permutations) — (HIS-RKHS) – KNN, dataset N4G4. Permutation p -value: <0.001 .

	Pred. Control	Pred. ADHD
True Control	185	115
True ADHD	75	230

SVM

Table 11: Repeated stratified subject-level evaluation (5 folds $\times$ 5 repeats = 25 train/test partitions), bootstrap 95% CI ($n = 2000$) — (HIS-RKHS) – SVM, dataset N4G4.

Metric	Mean	CI Lower	CI Upper
Accuracy	0.7254	0.6800	0.7658
Balanced Accuracy	0.7251	0.6799	0.7653
Sensitivity	0.7736	0.7205	0.8221
Specificity	0.6767	0.6233	0.7300
AUC-ROC	0.8126	0.7715	0.8506
MCC	0.4576	0.3661	0.5388

Table 12: Pooled confusion matrix over the 25 outer test folds and label-permutation test ($n = 1000$ permutations) — (HIS-RKHS) – SVM, dataset N4G4. Permutation p -value: <0.001 .

	Pred. Control	Pred. ADHD
True Control	203	97
True ADHD	69	236

Probability Product Kernel (PPK)

PPK-KNN

Table 13: Repeated stratified subject-level evaluation (5 folds $\times$ 5 repeats = 25 train/test partitions), bootstrap 95% CI ($n = 2000$) — Probability Product Kernel (PPK) – PPK-KNN, dataset N4G4.

Metric	Mean	CI Lower	CI Upper
Accuracy	0.7073	0.6755	0.7409
Balanced Accuracy	0.7060	0.6742	0.7395
Sensitivity	0.7587	0.7003	0.8095
Specificity	0.6533	0.5933	0.7133
AUC-ROC	0.7658	0.7309	0.7992
MCC	0.4279	0.3636	0.4961

Table 14: Pooled confusion matrix over the 25 outer test folds and label-permutation test ($n = 1000$ permutations) — Probability Product Kernel (PPK) – PPK-KNN, dataset N4G4. Permutation p -value: <0.001 .

	Pred. Control	Pred. ADHD
True Control	196	104
True ADHD	73	232

PPK-SVM

Table 15: Repeated stratified subject-level evaluation (5 folds $\times$ 5 repeats = 25 train/test partitions), bootstrap 95% CI ($n = 2000$) — Probability Product Kernel (PPK) – PPK-SVM, dataset N4G4.

Metric	Mean	CI Lower	CI Upper
Accuracy	0.7143	0.6806	0.7502
Balanced Accuracy	0.7135	0.6800	0.7492
Sensitivity	0.7803	0.7310	0.8272
Specificity	0.6467	0.5733	0.7133
AUC-ROC	0.7666	0.7228	0.8113
MCC	0.4422	0.3769	0.5122

Table 16: Pooled confusion matrix over the 25 outer test folds and label-permutation test ($n = 1000$ permutations) — Probability Product Kernel (PPK) – PPK-SVM, dataset N4G4. Permutation p -value: <0.001 .

	Pred. Control	Pred. ADHD
True Control	194	106
True ADHD	67	238

Dataset N5G5

(HIS-RKHS)

KNN

Table 17: Repeated stratified subject-level evaluation (5 folds $\times$ 5 repeats = 25 train/test partitions), bootstrap 95% CI ($n = 2000$) — (HIS-RKHS) – KNN, dataset N5G5.

Metric	Mean	CI Lower	CI Upper
Accuracy	0.7059	0.6723	0.7397
Balanced Accuracy	0.7050	0.6715	0.7390
Sensitivity	0.7633	0.7151	0.8090
Specificity	0.6467	0.5900	0.7000
AUC-ROC	0.7490	0.7101	0.7875
MCC	0.4207	0.3524	0.4895

Table 18: Pooled confusion matrix over the 25 outer test folds and label-permutation test ($n = 1000$ permutations) — (HIS-RKHS) – KNN, dataset N5G5. Permutation p -value: <0.001 .

	Pred. Control	Pred. ADHD
True Control	194	106
True ADHD	72	233

SVM

Table 19: Repeated stratified subject-level evaluation (5 folds $\times$ 5 repeats = 25 train/test partitions), bootstrap 95% CI ($n = 2000$) — (HIS-RKHS) – SVM, dataset N5G5. Only 5/25 individual fold outcomes were printed in the original log for this run (the remaining 20 folds were reloaded from a saved checkpoint during a resumed execution); the statistics below were computed by the pipeline over the complete set of 25 repeated stratified subject-level folds.

Metric	Mean	CI Lower	CI Upper
Accuracy	0.7119	0.6683	0.7534
Balanced Accuracy	0.7117	0.6679	0.7531
Sensitivity	0.7267	0.6784	0.7749
Specificity	0.6967	0.6300	0.7600
AUC-ROC	0.7988	0.7591	0.8371
MCC	0.4310	0.3428	0.5156

Table 20: Pooled confusion matrix over the 25 outer test folds and label-permutation test ($n = 1000$ permutations) — (HIS-RKHS) – SVM, dataset N5G5. Permutation p -value: <0.001 .

	Pred. Control	Pred. ADHD
True Control	209	91
True ADHD	83	222

Probability Product Kernel (PPK)

PPK-KNN

Table 21: Repeated stratified subject-level evaluation (5 folds $\times$ 5 repeats = 25 train/test partitions), bootstrap 95% CI ($n = 2000$) — Probability Product Kernel (PPK) – PPK-KNN, dataset N5G5.

Metric	Mean	CI Lower	CI Upper
Accuracy	0.7173	0.6919	0.7436
Balanced Accuracy	0.7162	0.6905	0.7426
Sensitivity	0.7656	0.7205	0.8095
Specificity	0.6667	0.6067	0.7300
AUC-ROC	0.7777	0.7494	0.8068
MCC	0.4499	0.3969	0.5025

Table 22: Pooled confusion matrix over the 25 outer test folds and label-permutation test ($n = 1000$ permutations) — Probability Product Kernel (PPK) – PPK-KNN, dataset N5G5. Permutation p -value: <0.001 .

	Pred. Control	Pred. ADHD
True Control	200	100
True ADHD	71	234

PPK-SVM

Table 23: Repeated stratified subject-level evaluation ($5 \text{ folds} \times 5 \text{ repeats} = 25 \text{ train/test partitions}$), bootstrap 95% CI ($n = 2000$) — Probability Product Kernel (PPK) – PPK-SVM, dataset N5G5.

Metric	Mean	CI Lower	CI Upper
Accuracy	0.7304	0.6905	0.7700
Balanced Accuracy	0.7304	0.6904	0.7700
Sensitivity	0.7774	0.7331	0.8231
Specificity	0.6833	0.6100	0.7567
AUC-ROC	0.8004	0.7558	0.8458
MCC	0.4712	0.3917	0.5529

Table 24: Pooled confusion matrix over the 25 outer test folds and label-permutation test ($n = 1000$ permutations) — Probability Product Kernel (PPK) – PPK-SVM, dataset N5G5. Permutation p -value: <0.001 .

	Pred. Control	Pred. ADHD
True Control	205	95
True ADHD	68	237

Table 25: Experiment 1 (nested cross-validation) — comparison across HMM topologies N3G3, N4G4, and N5G5 (number of hidden states $\times$ number of GMM components per state). Values are means over 25 outer folds (5×5 repeated nested CV); full bootstrap 95% confidence intervals and pooled confusion matrices for every entry are reported separately.

Configuration	Method	Accuracy	Bal. Acc.	Sens.	Spec.	AUC	MCC
N3G3							
	HIS-KNN	0.6642	0.6633	0.7067	0.6200	0.7314	0.3388
	HIS-SVM	0.7353	0.7350	0.7733	0.6967	0.7960	0.4828
	PPK-KNN	0.7235	0.7229	0.7992	0.6467	0.7649	0.4555
	PPK-SVM	0.7061	0.7055	0.7677	0.6433	0.7624	0.4245
N4G4							
	HIS-KNN	0.6855	0.6846	0.7526	0.6167	0.7642	0.3863
	HIS-SVM	0.7254	0.7251	0.7736	0.6767	0.8126	0.4576
	PPK-KNN	0.7073	0.7060	0.7587	0.6533	0.7658	0.4279
	PPK-SVM	0.7143	0.7135	0.7803	0.6467	0.7666	0.4422
N5G5							
	HIS-KNN	0.7059	0.7050	0.7633	0.6467	0.7490	0.4207
	HIS-SVM	0.7119	0.7117	0.7267	0.6967	0.7988	0.4310
	PPK-KNN	0.7173	0.7162	0.7656	0.6667	0.7777	0.4499
	PPK-SVM	0.7304	0.7304	0.7774	0.6833	0.8004	0.4712